Do Now · retrieval

Lesson 077 · Linear Search: Code and Trace

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Can linear search work with unsorted input?

Q2

What is not-found sentinel?

Q3

What does O(n) time mean?

Q4

Array a=[6,2,9,4]: set a[2]=a[0]+a[1]. Give updated array and total.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Visit each array value in order; linear search does not require sorted data.

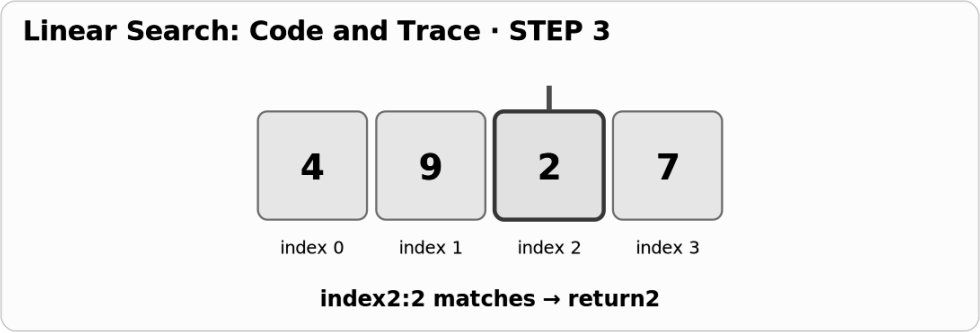
Core idea 2

Trace — Return the first matching index or a distinct not-found value such as −1.

Core idea 3

Explain — Worst-case O(n) time, O(1) extra storage; verify first, last, missing and empty inputs.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

linear search — Sequentially compare elements.

sentinel — Special result indicating a condition.

index — Zero-based position.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

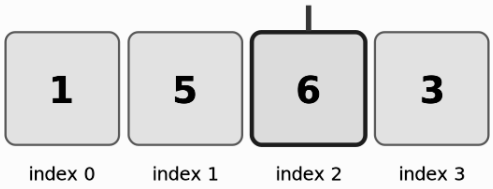
Find 6 in unsorted [1,5,6,3].

Follow the worked method

1. Check the working — Check indexes0,1,2, return2; stop before3.

Source model · final illustrated step

IDO · TRACE



Check indexes0,1,2, return2; stop before3.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Check the working
No equal item, so return−1 after four checks.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace 2 in [4,9,2,7], give checks and return. [6]

Question 2

Write C# linear search returning index/−1. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

State linear search best/worst time, space and one edge test. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Linear search requires a pre-sorted array.

Use key vocabulary from memory

1. Explain linear search in your own words.

2. Explain sentinel in your own words.

3. Explain index in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.